

EC400 – Static Optimization and Fixed Points
Problem Set 1: Tools for Optimization

1. Show that the general quadratic form

$$a_{11}x_1^2 + a_{12}x_1x_2 + a_{22}x_2^2$$

can be written as

$$\begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix};$$

and find its unique symmetric representation.

2. List all the principal minors of a general (3×3) matrix and denote which are the three leading principal minors.
3. Determine the definiteness of the following symmetric matrices:

$$(a) \begin{pmatrix} 0 & 0 \\ 0 & c \end{pmatrix}; \quad (b) \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}; \quad (c) \begin{pmatrix} -3 & 4 \\ 4 & -6 \end{pmatrix}; \quad (d) \begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 5 \\ 0 & 5 & 6 \end{pmatrix}.$$

4. [Harder] Consider the following quadratic form

$$Q(\mathbf{x}) = ax_1^2 + bx_2^2 + 2abx_1x_2.$$

For what values of the parameter values, a and b , is the quadratic form $Q(\mathbf{x})$ indefinite? Plot your answer in $\mathbb{R}^2$.

5. Approximate e^x at $x = 0$ with a Taylor polynomial of order three and four. Then compute the values of these approximation at $h = 0.2$ and at $h = 1$ and compare with the actual values.
6. For each of the following functions on $\mathbb{R}$, determine whether they are quasiconcave, quasiconvex, both, or neither:

$$(a) e^x; \quad (b) \ln x; \quad (c) x^3 - x$$

7. [Harder] Let f be a function defined on a convex set U in $\mathbb{R}^n$. In lecture, we have shown that f is a quasiconcave function on U if and only if for all $\mathbf{x}, \mathbf{y} \in U$ and $t \in [0, 1]$

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \geq \min\{f(\mathbf{x}), f(\mathbf{y})\}.$$

State the corresponding theorem for quasiconvex functions and prove it.

8. Prove that a weakly increasing transformation ($g : \mathbb{R} \rightarrow \mathbb{R}$ such that $g' \geq 0$) of a quasi-concave function is quasi-concave.
9. A commonly used production or utility function on $\mathbb{R}_+^2$ is $f(x, y) = xy$. Check whether it is concave or convex using its Hessian. Then, check whether it is quasiconcave.
10. [Harder] Show that any continuously differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$, satisfying

$$x \frac{\partial f(x)}{\partial x} \leq 0,$$

must be quasi-concave.

EC400 – Static Optimization and Fixed Points
Problem Set 2: Optimization

1. Find critical points, local maxima and local minima for each of the following functions:

- (a) $x^4 + x^2 - 6xy + 3y^2$
(b) $x^2 - 6xy + 2y^2 + 10x + 2y - 5$
(c) $xy^2 + x^3y - xy$

Which of the critical points are also global maxima or global minima?

2. Let $S \subset \mathbb{R}^n$ be an open set and $f : S \rightarrow \mathbb{R}$ be a twice continuously differentiable function. Suppose that $Df(x^*) = 0$. State the weakest sufficient conditions that the Hessian must satisfy at the critical point x^* for:

- (i) x^* to be a local max;
(ii) x^* to be a strict local min.

3. Check whether $f(x, y) = x^4 + x^2y^2 + y^4 - 3x - 8y$ is concave or convex by using the Hessian.

4. [Harder] Solve the following problem

$$\max_{x,y} [\min\{x, y\} - x^2 - y^2].$$

5. Find the optimal solution for the following program

$$\max_{x,y} x \quad \text{subject to} \quad x^3 + y^2 = 0.$$

Is the Lagrange approach appropriate?

6. Solve the following problem

$$\max_{x_1, x_2} x_1^2 x_2 \quad \text{subject to} \quad 2x_1^2 + x_2^2 = 3.$$

7. [Harder] Solve the following problem when $a \in [\frac{1}{2}, \frac{3}{2}]$

$$\max_{x,y \geq 0} x^2 + y^2 \quad \text{subject to} \quad ax + y = 1.$$

8. [Harder] Let X be a convex subset of $\mathbb{R}^n$, $f : X \rightarrow \mathbb{R}$ a concave function, $g : X \rightarrow \mathbb{R}^m$ a convex function, a is a vector in $\mathbb{R}^m$. Consider the following problem

$$\max_{x \in X} f(x) \quad \text{subject to} \quad g(x) \leq a.$$

What is the Lagrangian for this problem? Prove that the Lagrangian is a concave function of the choice variable x on X .

9. Consider the problem of maximizing xyz subject to $x + y + z \leq 1$, $x \geq 0$, $y \geq 0$, and $z \geq 0$. Obviously, the three latter constraints do not bind, and we can concentrate only on the first constraint, $x + y + z \leq 1$. Find the solution and the Lagrange multiplier, and show how the optimal value would change if instead the constraint was changed to $x + y + z \leq 9/10$.
10. [Harder] Consider a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfying:

$$f(x) = \begin{cases} u(x) & \text{if } g(x) \leq 0 \\ v(x) & \text{if } g(x) \geq 0 \end{cases}.$$

Further, suppose that: (i) $u(x) = v(x)$ if $g(x) = 0$; (ii) u and v are differentiable, strictly concave, and possess maximizers in $\mathbb{R}^n$; and (iii) $g(x)$ is differentiable and strictly convex. Carefully explain how you would solve the problem of maximizing f by choosing $x \in \mathbb{R}^n$.

EC400 – Static Optimization and Fixed Points
Problem Set 3: Comparative Statics and Correspondences

1. [Harder] For $a > 0$, consider the problem:

$$\begin{aligned} \max_{x,y} \quad & ax + y \\ \text{s.t.} \quad & x^2 + (x - y)^2 \leq 1 \\ & x \geq a, y \geq 0. \end{aligned}$$

Using the Kuhn-Tucker approach, write down the necessary first order conditions that must be satisfied by the solution of the constrained optimization problem. Are solutions to these conditions also maximizers of the Lagrangian? Solve the constrained optimization problem in terms of a . Find the slope of value function with respect to a , possibly relying on the Envelope Theorem.

2. Let a , x and y be non-negative. Consider the problem of maximizing xy subject to $x^2 + ay^2 \leq 1$. What happens to the optimal value when a increases marginally?
3. [Harder] Assume that the utility function of a consumer satisfies

$$u(x, y) = x + y^{1/2}.$$

The consumer has a positive income $I > 0$ and faces positive prices $p_x > 0$ and $p_y > 0$. The consumer cannot buy negative amounts of any of the goods.

- (a) Use Kuhn-Tucker to solve the consumer's problem.
- (b) Show how the utility at the maximum depends on I .
- (c) For the case of an interior solution, find how the endogenous variables change when I and p_x change. That is, compute:

$$\frac{\partial x}{\partial I}, \frac{\partial y}{\partial I}, \frac{\partial \mu_0}{\partial I}, \frac{\partial x}{\partial p_x}, \frac{\partial y}{\partial p_x}, \frac{\partial \mu_0}{\partial p_x}.$$

4. [Harder] Consider the system of equations:

$$\mathbf{F}(\mathbf{x}|b) = \begin{cases} x_1 - e^{x_2} \\ e^{x_1} + x_2 - b^2 \end{cases} = 0.$$

Argue whether you can invoke the implicit function theorem. Then, exploit the theorem where possible to find the slopes of the implicit functions $\mathbf{f}(b)$.

5. For each of the following correspondences $h : \mathbb{R} \rightrightarrows \mathbb{R}$ show whether they are convex-valued, upper-hemicontinuous or lower-hemicontinuous:

(a) $h(x) = [5x, 10x]$ for $x \in [0, 1]$;

(b) $h(x) = \{5x, 10x\}$ for $x \in [0, 1]$;

(c) $h(x) = \begin{cases} 1/2 & \text{for } x \in [0, 1) \\ [0, 1] & \text{for } x = 1 \end{cases}$.

6. Consider the correspondence $h : \mathbb{R} \rightrightarrows \mathbb{R}$ where

$$h(x) = \begin{cases} [0, 1/2] & \text{if } x > 1/3 \\ [1/2, 1] & \text{if } x < 1/3 \end{cases}.$$

Argue whether the Kakutani's Fixed Point Theorem applies and find all the fixed points.

7. Consider the correspondence $h : \mathbb{R}^2 \rightrightarrows \mathbb{R}^2$, where $\mathbf{h}(\mathbf{x}) = (h_1(\mathbf{x}), h_2(\mathbf{x}))$,

$$h_1(\mathbf{x}) = \begin{cases} 1 & \text{if } x_2 > 1/3 \\ [0, 1] & \text{if } x_2 = 1/3 \\ 0 & \text{if } x_2 < 1/3 \end{cases} \quad \text{and} \quad h_2(\mathbf{x}) = \begin{cases} 1 & \text{if } x_1 < 1/2 \\ [0, 1] & \text{if } x_1 = 1/2 \\ 0 & \text{if } x_1 > 1/2 \end{cases}.$$

Show how you can exploit Kakutani's Fixed Point Theorem despite the domain being open, and find all the fixed points.

8. [Harder] For all values of $b \geq 0$, solve the problem

$$\mathbf{x}^*(b) = \arg \max_{\mathbf{x} \in [0,1]^2} bx_1 + x_2 \quad \text{s.t.} \quad x_1 + x_2 = 1.$$

Show that $\mathbf{x}^*(b)$ is non-empty, compact-valued, convex-valued, and upper-hemicontinuous. Would this have followed also from the theorem of the maximum? Are the assumptions of the theorem satisfied?